Chapter 4: Motion in Two Dimensions

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4.1 The Position, Velocity, and Acceleration Vectors

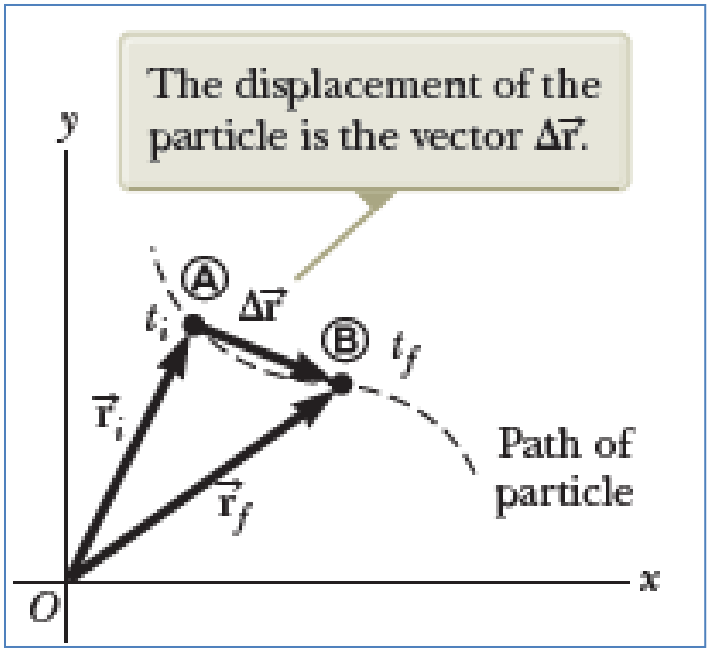


Figure 4.1 A particle moving in the *xy* plane is located with the position vector $\vec{r}$ drawn from the origin to the particle. The displacement of the particle as it moves from Ⓐ to Ⓑ in the time interval $\Delta t = t_f - t_i$ is equal to the vector $\Delta\vec{r} = \vec{r}_f - \vec{r}_i$.

$$\Delta\vec{r} \equiv \vec{r}_f - \vec{r}_i \qquad (4.1)$$

◀ Displacement vector

$$\vec{v}_{avg} \equiv \frac{\Delta\vec{r}}{\Delta t} \qquad (4.2)$$

◀ Average velocity

$$\vec{\mathbf{v}} \equiv \lim_{\Delta t \to 0} \frac{\Delta \vec{\mathbf{r}}}{\Delta t} = \frac{d\vec{\mathbf{r}}}{dt} \qquad (4.3)$$

◀ Instantaneous velocity

The magnitude of the instantaneous velocity vector $v = |\vec{\mathbf{v}}|$ of a particle is called the *speed* of the particle, which is a scalar quantity.

Average acceleration ▶

$$\vec{\mathbf{a}}_{\text{avg}} \equiv \frac{\Delta \vec{\mathbf{v}}}{\Delta t} = \frac{\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i}{t_f - t_i} \qquad (4.4)$$

Instantaneous acceleration ▶

$$\vec{\mathbf{a}} \equiv \lim_{\Delta t \to 0} \frac{\Delta \vec{\mathbf{v}}}{\Delta t} = \frac{d\vec{\mathbf{v}}}{dt} \qquad (4.5)$$

4.2 Two-Dimensional Motion with Constant Acceleration

motion in two dimensions can be modeled as two independent motions in each of the two perpendicular directions associated with the x and y axes. That is, any influence in the y direction does not affect the motion in the x direction and vice versa.

The position vector for a particle moving in the xy plane can be written

$$\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} \tag{4.6}$$

where x, y, and $\vec{\mathbf{r}}$ change with time as the particle moves while the unit vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ remain constant. If the position vector is known, the velocity of the particle can be obtained from Equations 4.3 and 4.6, which give

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} \tag{4.7}$$

Method I

$$in\ 1-D: v_{fx} = v_{ix} + a_x t \rightarrow but\ in\ 2-D: v_f = v_i + at = (v_{ix}i + v_{iy}j) + (a_x i + a_y j)t$$

$$in\ 1-D: x_f = x_i + v_{ix}t + \frac{1}{2}a_x t^2 \rightarrow$$

$$but\ in\ 2-D: r_f = r_i + v_i t + \frac{1}{2}at^2 = (x_i i + y_i j) + (v_{ix}i + v_{iy}j)t + \frac{1}{2}(a_x i + a_y j)t^2$$

Method II

$$\boldsymbol{v_f = v_{fx}i + v_{fy}j\ since\ v_{fx} = v_{ix} + a_x t\ and\ v_{fy} = v_{iy} + a_y t}$$

$$\boldsymbol{r_f = x_f i + y_f j\ since\ x_f = x_i + v_{ix}t + \frac{1}{2}a_x t^2\ and\ y_f = y_i + v_{iy}t + \frac{1}{2}a_y t^2}$$

Example 4.1 Motion in a Plane AM

A particle moves in the xy plane, starting from the origin at $t = 0$ with an initial velocity having an x component of 20 m/s and a y component of −15 m/s. The particle experiences an acceleration in the x direction, given by $a_x = 4.0\ \text{m/s}^2$.

(A) Determine the total velocity vector at any time.

$v_{xi} = 20\ \text{m/s}$, $v_{yi} = -15\ \text{m/s}$, $a_x = 4.0\ \text{m/s}^2$, and $a_y = 0$.

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = (v_{xi} + a_x t)\hat{\mathbf{i}} + (v_{yi} + a_y t)\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_f = [20 + (4.0)t]\hat{\mathbf{i}} + [-15 + (0)t]\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_f = [(20 + 4.0t)\hat{\mathbf{i}} - 15\hat{\mathbf{j}}]$$

In Method II:

$v_{fx} = v_{ix} + a_x t = 20 + 4t$ *and* $v_{fy} = v_{iy} + a_y t = -15 + 0 = -15$ *then*

$$v_f = v_{fx} i + v_{fy} j = (20 + 4t)i - 15j$$

(B) Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

$$\vec{\mathbf{v}}_f = [(20 + 4.0(5.0))\hat{\mathbf{i}} - 15\hat{\mathbf{j}}] = (40\hat{\mathbf{i}} - 15\hat{\mathbf{j}})\ \text{m/s}$$

$$\theta = \tan^{-1}\left(\frac{v_{yf}}{v_{xf}}\right) = \tan^{-1}\left(\frac{-15\ \text{m/s}}{40\ \text{m/s}}\right) = -21°$$

$$v_f = |\vec{\mathbf{v}}_f| = \sqrt{v_{xf}^2 + v_{yf}^2} = \sqrt{(40)^2 + (-15)^2}\ \text{m/s} = 43\ \text{m/s}$$

(C) Determine the x and y coordinates of the particle at any time t and its position vector at this time.

Use the components of Equation 4.9 with $x_i = y_i = 0$ at $t = 0$ and with x and y in meters and t in seconds:

$$x_f = v_{xi}t + \tfrac{1}{2}a_x t^2 = 20t + 2.0t^2$$

$$y_f = v_{yi}t = -15t$$

$$\vec{\mathbf{r}}_f = x_f\hat{\mathbf{i}} + y_f\hat{\mathbf{j}} = (20t + 2.0t^2)\hat{\mathbf{i}} - 15t\hat{\mathbf{j}}$$

4.3 Projectile Motion

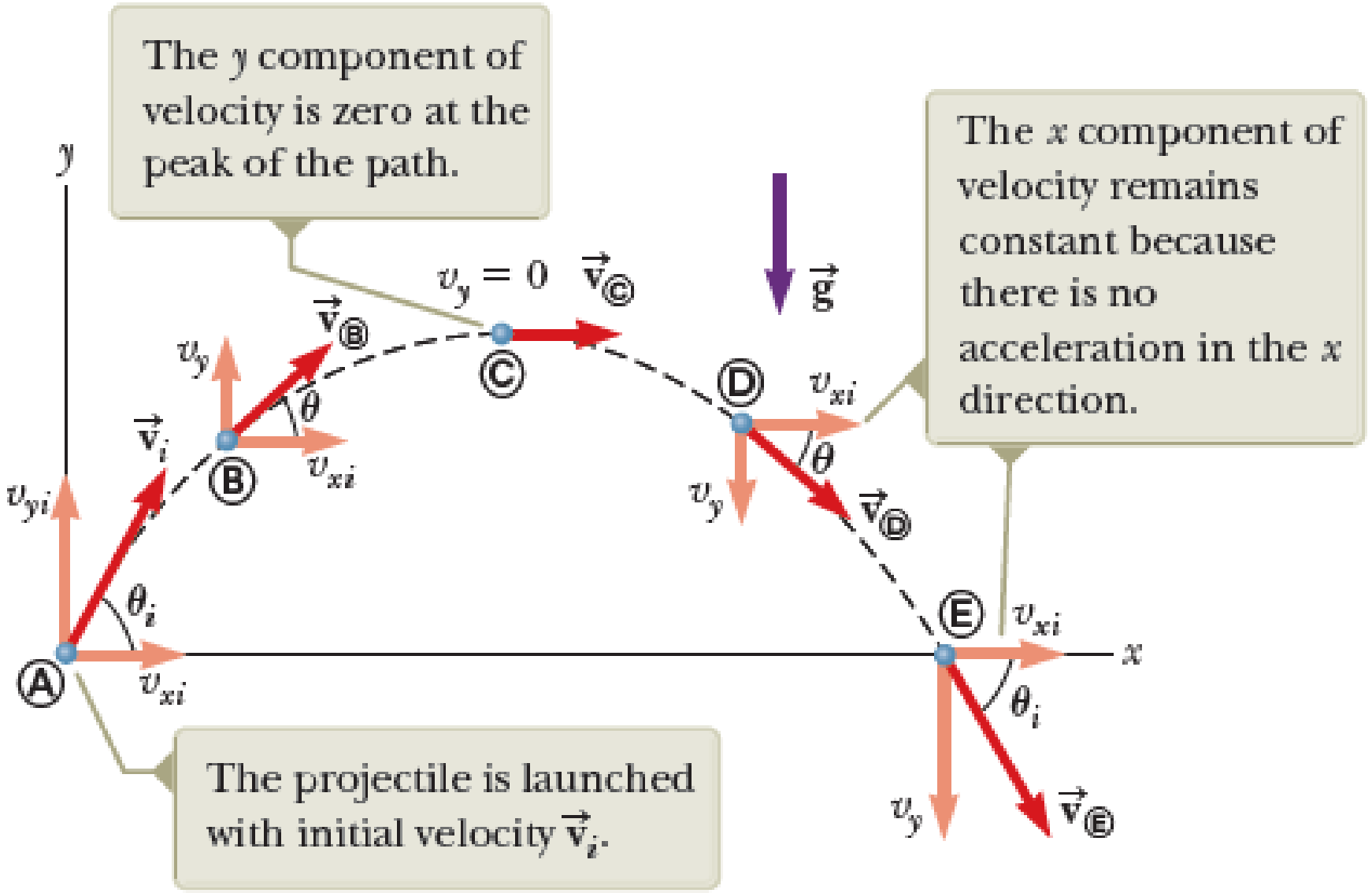
Projectile motion of an object is simple to analyze if we make two assumptions:

(1) the free-fall acceleration is constant over the range of motion and is directed downward, and

(2) the effect of air resistance is negligible.

With these assumptions, we find that the path of a projectile, which we call its *trajectory,* ***is always a parabola as shown in Figure 4.7.***

We use these assumptions throughout this chapter.



$$v_{xi} = v_i \cos \theta_i \qquad v_{yi} = v_i \sin \theta_i$$

$$x_f = x_i + v_{xi}t$$

$$v_{yf} = v_{yi} - gt$$

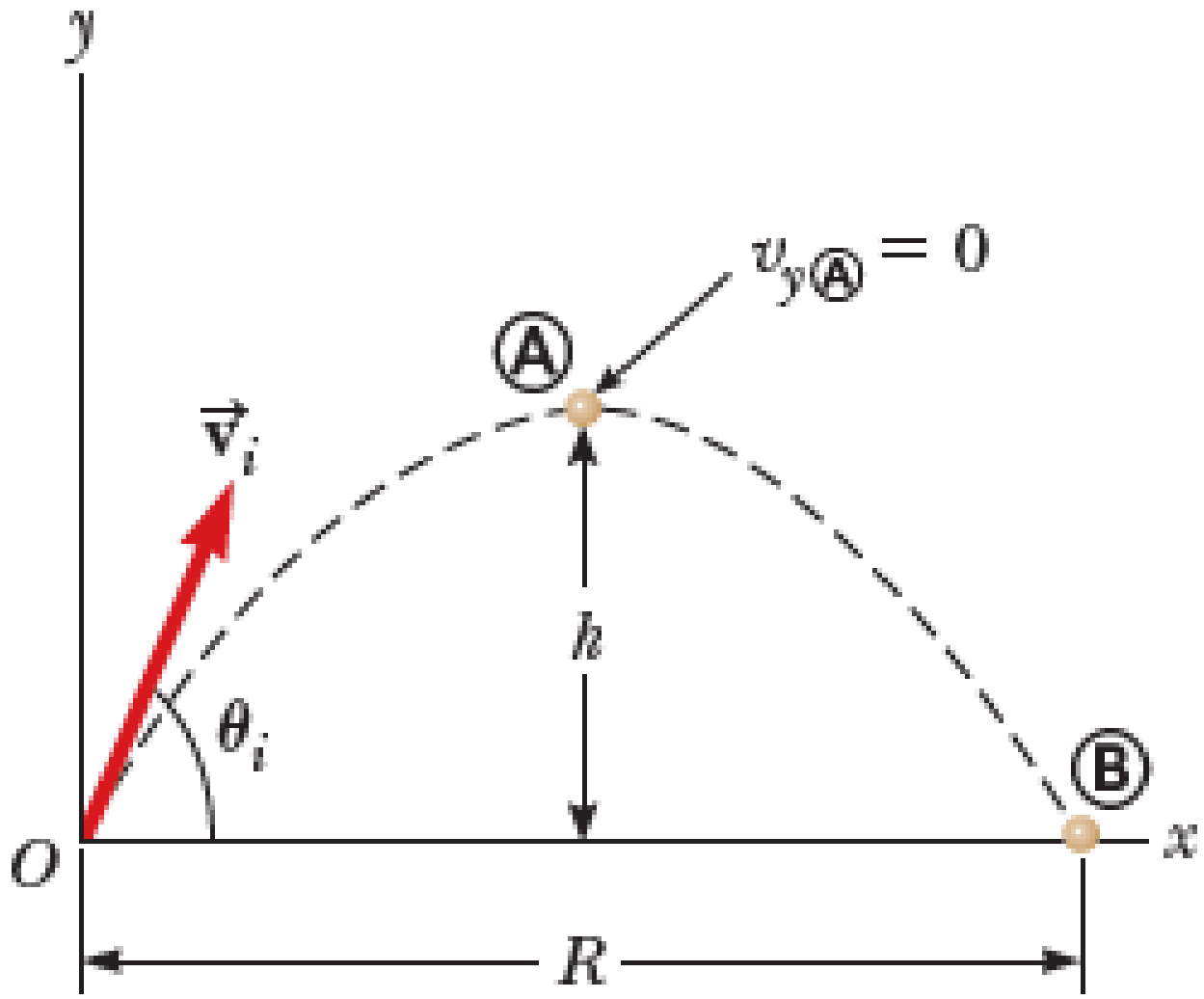
$$v_{y,\text{avg}} = \frac{v_{yi} + v_{yf}}{2}$$

$$y_f = y_i + \tfrac{1}{2}(v_{yi} + v_{yf})t$$

$$y_f = y_i + v_{yi}t - \tfrac{1}{2}gt^2$$

$$v_{yf}^2 = v_{yi}^2 - 2g(y_f - y_i)$$

Horizontal Range (R) and Maximum Height (h) of a Projectile



We can determine h by noting that at the peak $v_{y\text{Ⓐ}} = 0$. Therefore, from the particle under constant acceleration model, we can use the y direction version of Equation 2.13 to determine the time $t_{\text{Ⓐ}}$ at which the projectile reaches the peak:

$$v_{yf} = v_{yi} - gt \quad \rightarrow \quad 0 = v_i \sin\theta_i - gt_{\text{Ⓐ}}$$

$$t_{\text{Ⓐ}} = \frac{v_i \sin\theta_i}{g}$$

Substituting this expression for $t_{\text{Ⓐ}}$ into the y direction version of Equation 2.16 and replacing $y_f = y_{\text{Ⓐ}}$ with h, we obtain an expression for h in terms of the magnitude and direction of the initial velocity vector:

$$y_f = y_i + v_{yi}t - \tfrac{1}{2}gt^2 \quad \rightarrow \quad h = (v_i \sin\theta_i)\frac{v_i \sin\theta_i}{g} - \tfrac{1}{2}g\left(\frac{v_i \sin\theta_i}{g}\right)^2$$

$$h = \frac{v_i^2 \sin^2\theta_i}{2g} \tag{4.12}$$

$$h = \frac{v_i^2 \sin^2\theta_i}{2g}$$

The range R is the horizontal position of the projectile at a time that is twice the time at which it reaches its peak, that is, at time $t_B = 2t_A$. Using the particle under constant velocity model, noting that $v_{xi} = v_{xB} = v_i \cos\theta_i$, and setting $x_B = R$ at $t = 2t_A$, we find that

$$x_f = x_i + v_{xi}t \rightarrow R = v_{xi}t_B = (v_i \cos\theta_i)2t_A$$

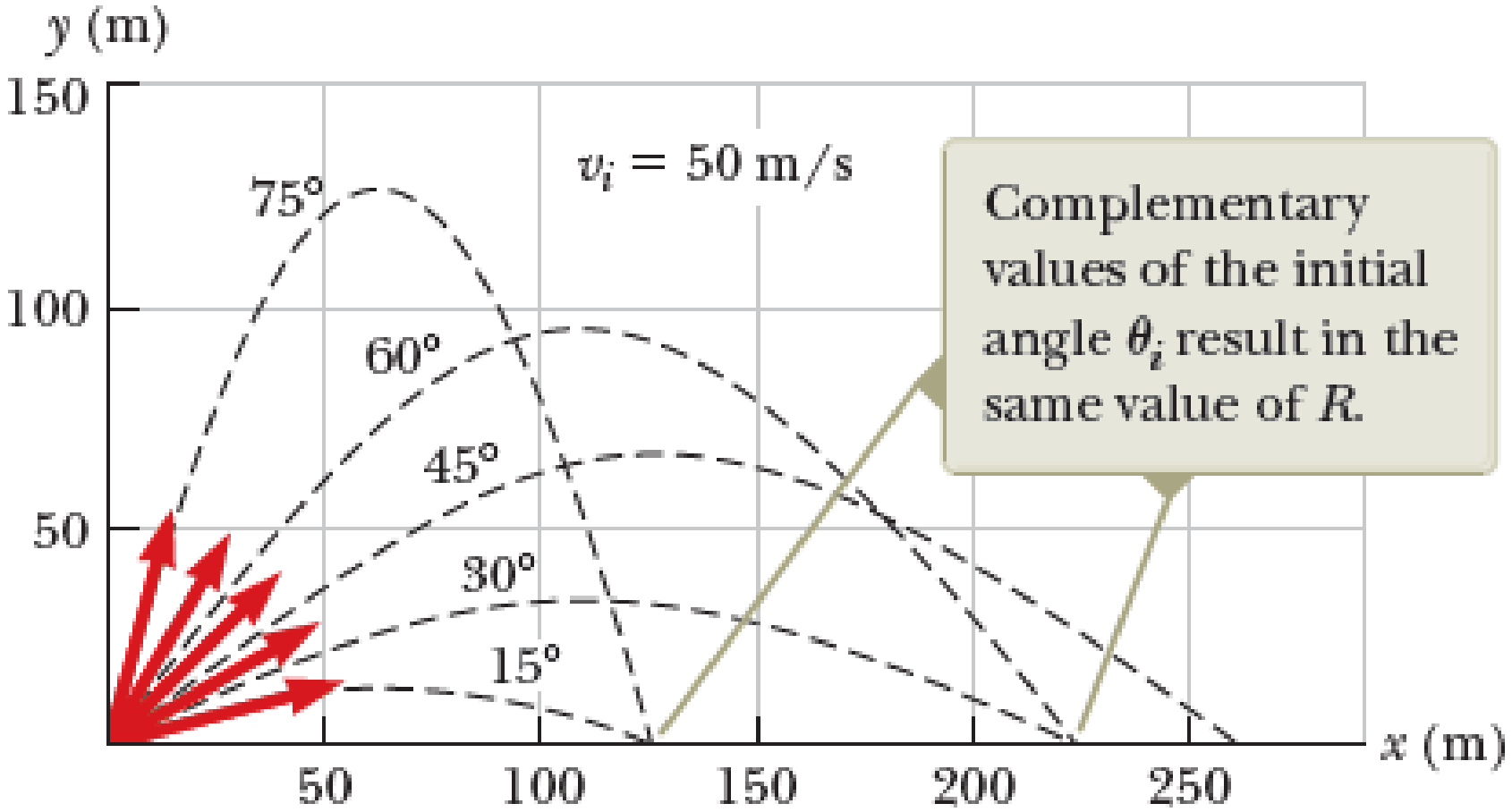
$$= (v_i \cos\theta_i)\frac{2v_i \sin\theta_i}{g} = \frac{2v_i^2 \sin\theta_i \cos\theta_i}{g}$$

Using the identity $\sin 2\theta = 2 \sin\theta \cos\theta$ (see Appendix B.4), we can write R in the more compact form

$$R = \frac{v_i^2 \sin 2\theta_i}{g} \tag{4.13}$$

The maximum value of R from Equation 4.13 is $R_{max} = v_i^2/g$. This result makes sense because the maximum value of $\sin 2\theta_i$ is 1, which occurs when $2\theta_i = 90°$. Therefore, R is a maximum when $\theta_i = 45°$.

$$R = \frac{v_i^2 \sin 2\theta_i}{g}$$


y (m)
150
100
50
$v_i = 50$ m/s
75°
60°
45°
30°
15°
Complementary values of the initial angle θ_i result in the same value of R.
x (m)
50
100
150
200
250

Example 4.2 The Long Jump

A long jumper (Fig. 4.11) leaves the ground at an angle of 20.0° above the horizontal and at a speed of 11.0 m/s.

(A) How far does he jump in the horizontal direction?

$$R = \frac{v_i^2 \sin 2\theta_i}{g} = \frac{(11.0\ \text{m/s})^2 \sin 2(20.0^\circ)}{9.80\ \text{m/s}^2} = 7.94\ \text{m}$$



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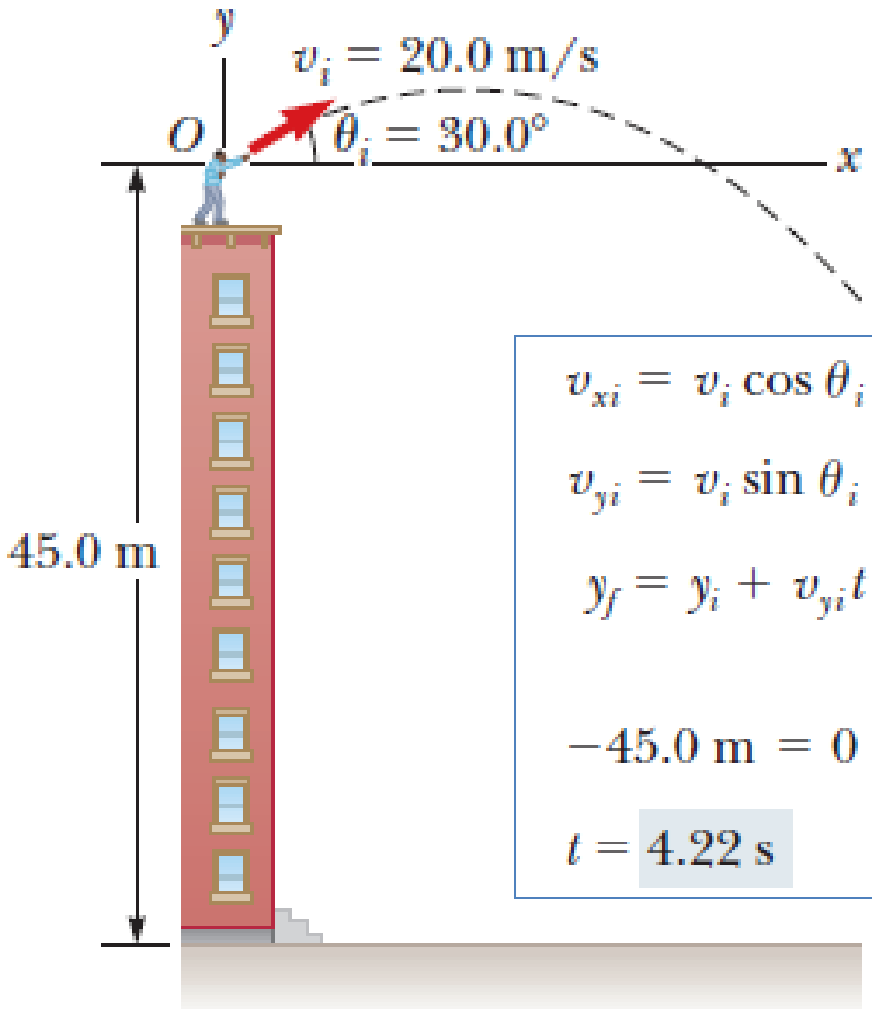
(B) What is the maximum height reached?

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g} = \frac{(11.0\ \text{m/s})^2 (\sin 20.0^\circ)^2}{2(9.80\ \text{m/s}^2)} = 0.722\ \text{m}$$

Example 4.4 That's Quite an Arm! AM

A stone is thrown from the top of a building upward at an angle of 30.0° to the horizontal with an initial speed of 20.0 m/s as shown in Figure 4.13. The height from which the stone is thrown is 45.0 m above the ground.

(A) How long does it take the stone to reach the ground?



$$v_{xi} = v_i \cos\theta_i = (20.0 \text{ m/s}) \cos 30.0° = 17.3 \text{ m/s}$$

$$v_{yi} = v_i \sin\theta_i = (20.0 \text{ m/s}) \sin 30.0° = 10.0 \text{ m/s}$$

$$y_f = y_i + v_{yi}t - \tfrac{1}{2}gt^2$$

$$-45.0 \text{ m} = 0 + (10.0 \text{ m/s})t + \tfrac{1}{2}(-9.80 \text{ m/s}^2)t^2$$

$$t = 4.22 \text{ s}$$

(B) What is the speed of the stone just before it strikes the ground?

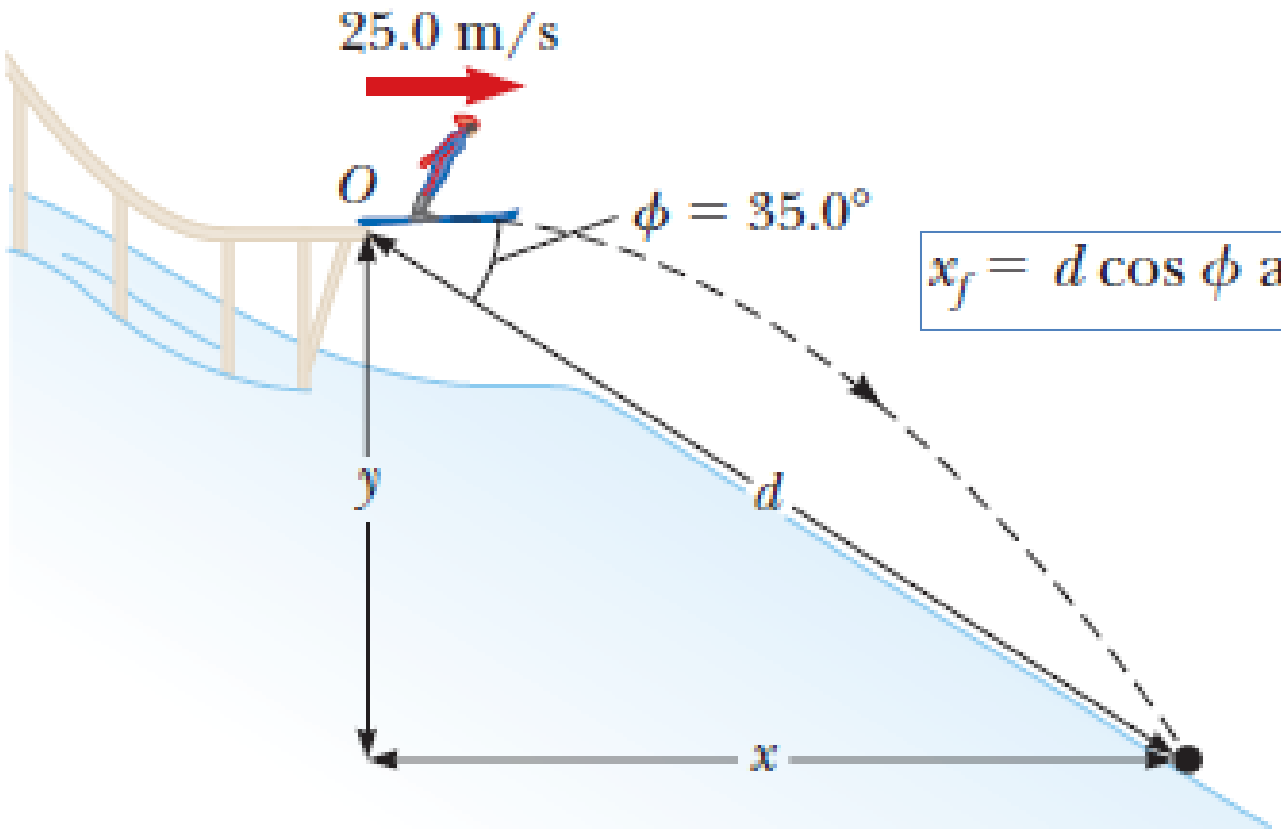
$$v_{yf} = v_{yi} - gt$$

$$v_{yf} = 10.0\ \text{m/s} + (-9.80\ \text{m/s}^2)(4.22\ \text{s}) = -31.3\ \text{m/s}$$

$$v_f = \sqrt{v_{xf}^2 + v_{yf}^2} = \sqrt{(17.3\ \text{m/s})^2 + (-31.3\ \text{m/s})^2} = 35.8\ \text{m/s}$$

Example 4.5 The End of the Ski Jump AM

A ski jumper leaves the ski track moving in the horizontal direction with a speed of 25.0 m/s as shown in Figure 4.14. The landing incline below her falls off with a slope of 35.0°. Where does she land on the incline?



$$x_f = d\cos\phi \text{ and } y_f = -d\sin\phi.$$

Figure 4.14 (Example 4.5) A ski jumper leaves the track moving in a horizontal direction.

$$x_f = d\cos\phi \text{ and } y_f = -d\sin\phi.$$

(1) $x_f = v_{xi}t$

(2) $y_f = v_{yi}t - \frac{1}{2}gt^2$

(3) $d\cos\phi = v_{xi}t$

(4) $-d\sin\phi = -\frac{1}{2}gt^2$

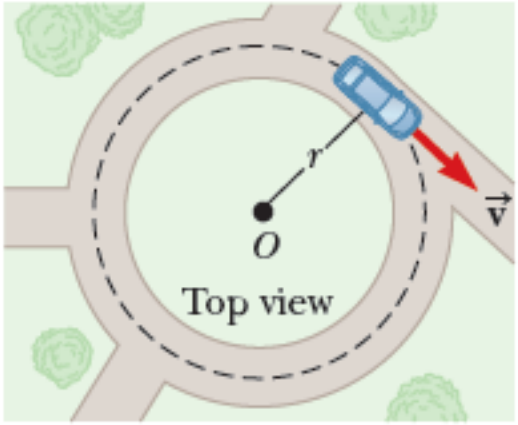
$$-d\sin\phi = -\frac{1}{2}g\left(\frac{d\cos\phi}{v_{xi}}\right)^2$$

$$d = \frac{2v_{xi}^2\sin\phi}{g\cos^2\phi} = \frac{2(25.0\text{ m/s})^2\sin 35.0^\circ}{(9.80\text{ m/s}^2)\cos^2 35.0^\circ} = 109\text{ m}$$

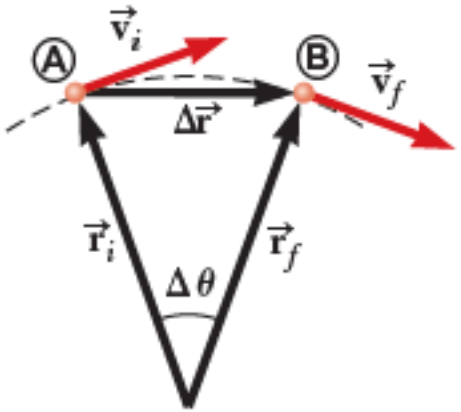
$$x_f = d\cos\phi = (109\text{ m})\cos 35.0^\circ = 89.3\text{ m}$$

$$y_f = -d\sin\phi = -(109\text{ m})\sin 35.0^\circ = -62.5\text{ m}$$

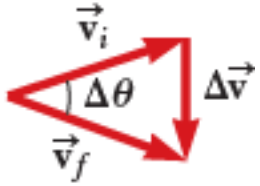
4.4 Analysis Model: Particle in Uniform Circular Motion



a



b



c

$$\frac{|\Delta\vec{v}|}{v} = \frac{|\Delta\vec{r}|}{r}$$

$$|\vec{a}_{avg}| = \frac{|\Delta\vec{v}|}{|\Delta t|} = \frac{v|\Delta\vec{r}|}{r\Delta t}$$

Centripetal acceleration for a particle in uniform circular motion

$$a_c = \frac{v^2}{r}$$

Period of circular motion for a particle in uniform circular motion

$$T = \frac{2\pi r}{v}$$

angular speed ω

$$\omega = \frac{2\pi}{T}$$

$$\omega = 2\pi\left(\frac{v}{2\pi r}\right) = \frac{v}{r} \quad \rightarrow \quad v = r\omega$$

$$a_c = \frac{(r\omega)^2}{r}$$

$$a_c = r\omega^2$$

Example 4.6 The Centripetal Acceleration of the Earth AM

(A) What is the centripetal acceleration of the Earth as it moves in its orbit around the Sun?

the radius of the Earth's orbit around the Sun, which is 1.496×10^{11} m.

$$a_c = \frac{v^2}{r} = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{4\pi^2 r}{T^2}$$

$$a_c = \frac{4\pi^2(1.496 \times 10^{11}\ \text{m})}{(1\ \text{yr})^2}\left(\frac{1\ \text{yr}}{3.156 \times 10^7\ \text{s}}\right)^2 = 5.93 \times 10^{-3}\ \text{m/s}^2$$

(B) What is the angular speed of the Earth in its orbit around the Sun?

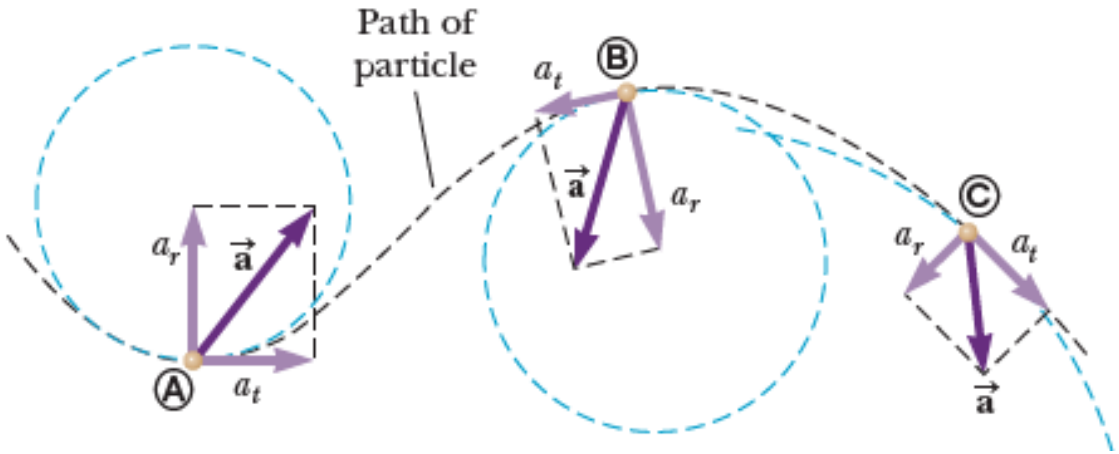
$$\omega = \frac{2\pi}{1\ \text{yr}}\left(\frac{1\ \text{yr}}{3.156 \times 10^7\ \text{s}}\right) = 1.99 \times 10^{-7}\ \text{s}^{-1}$$

4.5 Tangential and Radial Acceleration

$$\vec{\mathbf{a}} = \vec{\mathbf{a}}_r + \vec{\mathbf{a}}_t \tag{4.19}$$

The tangential acceleration component causes a change in the speed v of the particle. This component is parallel to the instantaneous velocity, and its magnitude is given by

$$a_t = \left| \frac{dv}{dt} \right| \tag{4.20}$$



$$a_r = -a_c = -\frac{v^2}{r} \tag{4.21}$$

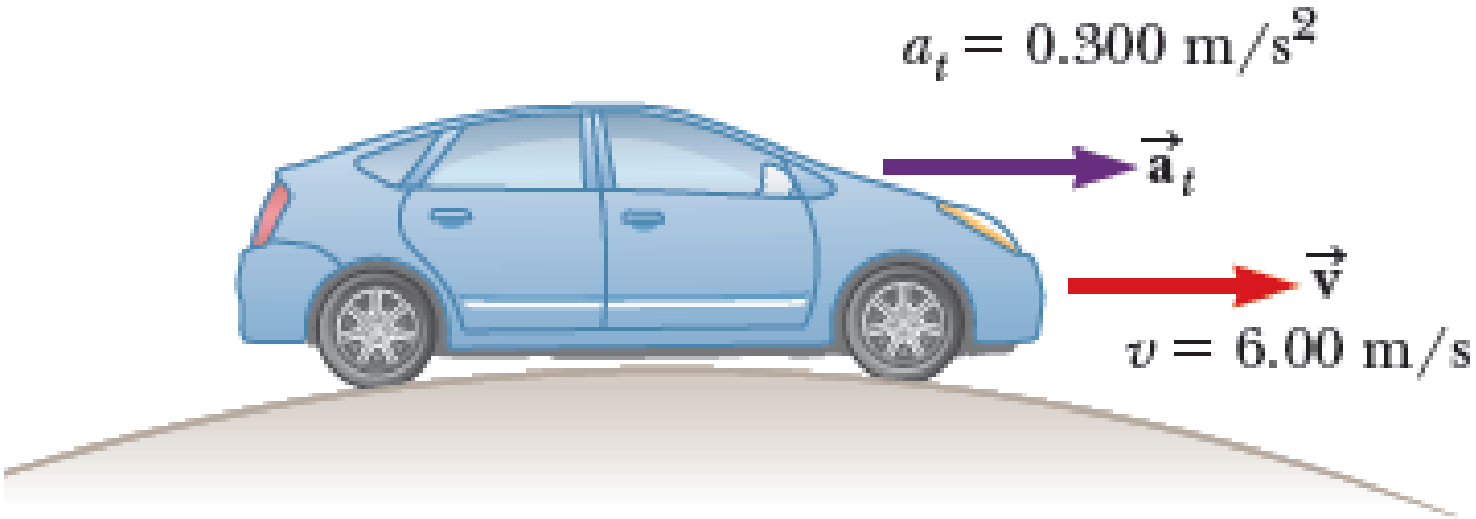
◀ Radial acceleration

$$a = \sqrt{a_r^2 + a_t^2}.$$

$$\phi = \tan^{-1} \frac{a_r}{a_t}$$

Example 4.7 Over the Rise

A car leaves a stop sign and exhibits a constant acceleration of $0.300\ \text{m/s}^2$ parallel to the roadway. The car passes over a rise in the roadway such that the top of the rise is shaped like an arc of a circle of radius 500 m. At the moment the car is at the top of the rise, its velocity vector is horizontal and has a magnitude of 6.00 m/s. What are the magnitude and direction of the total acceleration vector for the car at this instant?



$$a_r = -\frac{v^2}{r} = -\frac{(6.00\ \text{m/s})^2}{500\ \text{m}} = -0.072\ 0\ \text{m/s}^2$$

$$\sqrt{a_r^2 + a_t^2} = \sqrt{(-0.072\ 0\ \text{m/s}^2)^2 + (0.300\ \text{m/s}^2)^2}$$

$$= 0.309\ \text{m/s}^2$$

$$\phi = \tan^{-1}\frac{a_r}{a_t} = \tan^{-1}\left(\frac{-0.072\ 0\ \text{m/s}^2}{0.300\ \text{m/s}^2}\right) = -13.5^\circ$$